Phase 1: Problem Statement Document

Project: Pharma-Pulse: Generic Drug Shortage Intelligence System

Author: Harekrishna Thakur **Date:** April 2027 **Version:** 1.0

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1. Problem Background

India's public health system serves over 800 million beneficiaries through government-run hospitals, district health centres, and state procurement agencies. These institutions depend on a continuous supply of WHO essential generic medicines for routine and emergency care.

Currently, shortage detection across this system is entirely reactive. Procurement officers identify supply gaps only when physical stock is depleted, at which point the standard procurement and supply lead time of 8 to 12 weeks makes a timely recovery operationally impossible. There exists no early warning mechanism, no cross-signal intelligence layer, and no decision-support tool accessible to district or state-level procurement officers.

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2. Problem Statement

Public health procurement bodies in India lack a predictive intelligence system capable of identifying generic drug shortage risk 8–12 weeks before onset, resulting in stock-outs that directly compromise patient care continuity in government health facilities.

3. Who is Affected?

Stakeholder	How They Are Affected
District Procurement Officers	Make reactive decisions with no forward visibility; manage 300+ drugs manually
State Health Agencies (e.g. TNMSC, MSMSC)	Cannot pre-empt tender gaps or buffer stock requirements
Hospital Pharmacists	First to detect shortage — no upstream warning reaches them
Clinicians	Forced to alter treatment protocols when drugs are unavailable
Patients	Experience treatment delays, medication gaps, adverse health outcomes

4. Root Causes Identified

Root Cause 1: Data fragmentation. Shortage-relevant signals exist across at least five agencies , CDSCO, GeM, IDSP, state health departments, and commodity price indices , with no integration layer connecting them. Each institution sees one piece of the picture. Nobody sees the whole.

Root Cause 2: No defined intelligence ownership. The early warning gap sits among the manufacturer, distributor, and procurement body. It is owned by none of them. Each waits for the other to raise the alarm.

Root Cause 3: Commercial neglect of the public sector. Private sector analytics tools serve large pharmaceutical portfolios and private hospital chains. Public health procurement in tier-2 and tier-3 India presents no profitable commercial opportunity. No company has built for this segment. The gap remains unaddressed.

Root Cause 4: Exclusively backwards-looking reporting. All current government health reporting is historical. No institution publishes or acts on forward-looking shortage risk indicators. The system is structurally designed to react, not anticipate.

5. Evidence

Event	Year	Drugs Affected	Impact	Upstream Signal Available?
COVID-19 second wave demand surge	2021	Paracetamol IV, Azithromycin, Remdesivir	Critical stock-outs across 14 states during peak clinical demand	Yes — demand spike pattern and API price movement visible 10 weeks prior in public data
Chinese API export restriction impact	2023	Amoxicillin 500mg	Shortage reported across 6 state procurement systems	Yes — Chinese manufacturing disruption and export volume decline visible 8 weeks prior
Oncology generic supply gap	2022	Methotrexate (generic)	3-week supply gap across 4 major government cancer centres	Yes — CDSCO regulatory approval backlog flagged 12 weeks prior
West Asia conflict — API supply disruption	2026	Paracetamol, Amoxicillin, Metformin, Azithromycin	National shortage warning issued by FOPE; emergency escalation to Department of Pharmaceuticals, Ministry of Chemicals and Fertilisers	Yes — geopolitical disruption signal and API price surge visible upstream 8-10 weeks prior

Note on the 2026 event: The Federation of Pharma Entrepreneurs formally warned the Department of Pharmaceuticals of an imminent shortage of four essential generic medicines, attributing the crisis directly to API scarcity driven by the West Asia conflict. This event validates Pharma-Pulse's core architecture; the upstream signals that caused this shortage are precisely the variables our model monitors. The crisis became publicly visible only after prices had already surged. Pharma-Pulse is designed to surface this risk before that inflexion point.

Source: Bajeli-Datt, K. (2026, March 17). The New Indian Express.

6. Consequence of Inaction

Without an early warning system, the current pattern continues, and shortages occur on average 9 weeks after actionable signals were available in public data, patients experience medication gaps, and procurement officers continue operating reactively with no decision support.

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7. Proposed Solution Direction

Development of a BA-led intelligence system: Pharma-Pulse, that aggregates multi-source public data, applies a weighted risk scoring model, and delivers a procurement decision dashboard providing 8–12 weeks of forward warning on generic drug shortage risk across Indian states.

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8. What Success Looks Like

A district procurement officer receives an automated risk alert 10 weeks before a projected shortage. They initiate a pre-emptive tender. The drug remains available. The patient experiences no interruption in treatment.
